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Assignment - Cloud Security Essentials

Module 1 – Fundamentals

1. What is Cloud Computing?

Answer - Cloud computing is a technology that allows users to access computing resources such as servers, storage, databases, networking, and software through the internet instead of using local computers or physical servers. These resources are provided by cloud service providers and can be accessed from anywhere.

Cloud computing helps organizations reduce hardware costs, improve scalability, and increase efficiency. Users only pay for the services they use, making cloud computing a cost-effective solution for businesses and individuals.

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2. Describe Cloud Computing Deployment Model.

Answer - Cloud deployment models define how cloud infrastructure is deployed and who has access to it.

1. Public Cloud

Public cloud services are provided by third-party providers over the internet. These resources are shared among multiple users.

Examples: AWS, Google Cloud, Microsoft Azure.

2. Private Cloud

A private cloud is dedicated to a single organization. It provides more security and control.

Example: Internal company cloud infrastructure.

3. Hybrid Cloud

Hybrid cloud combines both public and private cloud environments. Organizations can store sensitive data in a private cloud and other data in a public cloud.

4. Community Cloud

This cloud infrastructure is shared by multiple organizations with similar requirements such as government departments or research institutions.

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3. Describe Different Types of Cloud Services

Answer - 1. Infrastructure as a Service (IaaS)

Provides virtualized computing resources such as servers, storage, and networking.

Example: AWS EC2.

2. Platform as a Service (PaaS)

Provides a platform for developers to build, test, and deploy applications without managing infrastructure.

Example: Google App Engine.

3. Software as a Service (SaaS)

Provides software applications over the internet.

Example: Gmail, Microsoft Office 365.

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4. Describe Cloud Computing Architecture

Answer - Cloud computing architecture consists of different components that work together to deliver cloud services.

Front End

This is the client side where users interact with the cloud through browsers or applications.

Back End

The back end includes servers, storage systems, databases, and cloud management software.

Network

The internet acts as the communication medium between the front end and back end.

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5. Components of Cloud Computing

Answer - The main components of cloud computing are:

Client Infrastructure – Devices used by users to access the cloud.

Application – Software delivered through the cloud.

Service – Services like IaaS, PaaS, and SaaS.

Runtime Cloud – Execution environment for applications.

Storage – Data storage systems used in cloud.

Infrastructure – Physical hardware like servers and networking.

Management – Tools used to manage cloud resources.

Security – Mechanisms to protect cloud data and services.

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6. Advantages and Disadvantages of Cloud Computing

Answer –

Advantages

- **Cost saving (no hardware purchase required)**
- **Scalability and flexibility**
- **Easy accessibility from anywhere**
- **Automatic updates and maintenance**
- **High storage capacity**

Disadvantages

- **Internet dependency**
- **Security and privacy concerns**
- **Limited control over infrastructure**
- **Possible downtime if service provider fails**

Module 2 – Virtualization and Storage Management

1. What is Virtualization and its Types?

Answer - Virtualization is the technology that allows multiple virtual machines to run on a single physical server. It helps optimize hardware resources.

Types of Virtualization

- **Server Virtualization – Multiple virtual servers on one physical server.**
- **Storage Virtualization – Combines multiple storage devices into a single storage system.**
- **Network Virtualization – Creates virtual networks.**
- **Desktop Virtualization – Allows users to access desktops remotely.**

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2. Types of Hypervisor and How to Manage It

Answer - A hypervisor is software that creates and manages virtual machines.

Type 1 Hypervisor (Bare Metal)

Runs directly on hardware.

Examples: VMware ESXi, Microsoft Hyper-V.

Type 2 Hypervisor (Hosted)

Runs on top of an operating system.

Examples: VirtualBox, VMware Workstation.

Management is done using hypervisor management tools such as vSphere, Hyper-V Manager, or command line tools.

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3. What is Snapshot and Cloning?

Answer - Snapshot

A snapshot captures the current state of a virtual machine including memory, disk, and settings. It allows users to revert the system to that state later.

Cloning

Cloning creates an exact copy of a virtual machine. It is used to quickly deploy multiple identical systems.

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4. Role of Virtualization in Cloud Computing

Answer - Virtualization is the core technology behind cloud computing.

It helps:

- **Efficient use of hardware**
- **Faster resource allocation**
- **Easy scalability**
- **Isolation between users**
- **Quick deployment of services**

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5. What is a Container?

Answer - A container is a lightweight virtualization technology that packages an application and its dependencies together

so it can run consistently on different systems.

Containers share the host operating system but run in isolated environments.

Example: Docker containers.

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6. High Availability and Live Migration

Answer –

High Availability (HA)

High availability ensures that services remain available even if a server fails by automatically switching to another server.

Live Migration

Live migration allows a virtual machine to move from one physical server to another without shutting down the system.

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7. Storage Configuration

Answer - Block Storage

Data is stored in fixed-size blocks. Used for databases and operating systems.

File Storage

Data is stored in files and directories.

Object Storage

Stores data as objects with metadata. Used in cloud platforms.

DAS (Direct Attached Storage)

Storage directly connected to a computer.

NAS (Network Attached Storage)

Storage connected to a network and accessed by multiple users.

SAN (Storage Area Network)

High-speed network that connects servers to storage devices.

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8. Storage Allocation and Provisioning

Answer - Storage allocation means assigning storage resources to users or applications.

Provisioning is the process of creating and managing storage resources automatically when needed.

Types include:

- **Thin provisioning**
- **Thick provisioning**
- **Dynamic provisioning**

Module 3 – System Management and Public Cloud

1. **Different Types of Cloud Storage**

Answer -

- **Object Storage**
- **Block Storage**
- **File Storage**
- **Archive Storage**
- **Backup Storage**

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2. **Role Based Access Control (RBAC), IAM and MFA**

Answer -

RBAC

Access permissions are assigned based on user roles.

IAM (Identity and Access Management)

Manages user identities and controls access to resources.

MFA (Multi-Factor Authentication)

Adds additional security by requiring multiple verification methods.

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3. Physical and Virtual Host Allocation

Answer - Physical host allocation assigns workloads to physical servers.

Virtual host allocation assigns workloads to virtual machines running on physical servers.

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4. Accessing Cloud Resources

Answer - Cloud resources can be accessed using:

- **Web dashboard**
- **Command line interface**
- **APIs**
- **Remote access tools**

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5. Types of Backup in Cloud

Answer -

- **Full Backup**
- **Incremental Backup**
- **Differential Backup**
- **Snapshot Backup**

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6. Disaster Recovery

Answer - Disaster recovery is a strategy used to restore systems and data after

failures, natural disasters, or cyber attacks.

It ensures business operations continue with minimal downtime.

Module 4 – Resource Management and Security

1. Resource Monitoring Techniques

Answer -

Resource monitoring tracks system performance and usage.

Tools include:

- **CloudWatch**
- **Azure Monitor**
- **Google Cloud Monitoring**

Metrics monitored include CPU usage, memory usage, network traffic, and storage usage.

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2. Accessing Compute from Internet

Answer - Users can access Windows or Linux cloud servers using remote access tools.

Tools

- **SSH (Secure Shell) – Linux servers**
- **RDP (Remote Desktop Protocol) – Windows servers**

Security methods include firewalls, VPN, and authentication keys.

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3. Encryption Technologies and Methods

Answer - Encryption protects data by converting it into unreadable form.

Types include:

- **Data at rest encryption**
- **Data in transit encryption**
- **End-to-end encryption**

Common algorithms include AES and RSA.

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4. Cloud Security

Answer –

Network Security

-Protects cloud networks using firewalls, VPN, and network segmentation.

Compute Security

-Protects virtual machines and operating systems.

Storage Security

-Ensures data protection using encryption and access control.

Module 5 – Business Continuity in the Cloud

1. Security and Privacy in Cloud

Answer - Security and privacy are maintained using:

- **Encryption**
- **Identity management**
- **Security monitoring**
- **Compliance policies**

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2. Portability in Cloud

Answer - Portability means the ability to move applications and data between different cloud platforms without major modifications.

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3. Reliability and High Availability

Answer - Reliability ensures that cloud services operate consistently without failure.

High availability ensures that services remain accessible even during hardware or system failures.

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4. Mobile Cloud Computing

Answer - Mobile cloud computing integrates cloud computing with mobile devices, allowing users to access cloud services through smartphones and tablets.

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5. AWS, Azure and Google Cloud Platforms

Answer –

AWS

-Amazon Web Services provides scalable cloud services including computing, storage, and databases.

Azure

-Microsoft Azure offers cloud services integrated with Microsoft products.

Google Cloud

-Google Cloud Platform provides cloud computing services focused on data analytics and machine learning.

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6. Accessing AWS, Azure and Google Cloud

Answer - Users can access these platforms through:

- **Web console**
- **CLI tools**
- **APIs**
- **SDKs**

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7. Creating Compute, Network and Storage

Answer –

Compute

-Create virtual machines or instances.

Network

-Configure virtual networks and security groups.

Storage

-Create storage buckets or disks.

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8. Compare Cloud Pricing

Answer - Cloud pricing depends on:

- **Compute usage**
- **Storage capacity**
- **Data transfer**
- **Additional services**

AWS, Azure, and Google Cloud follow a pay-as-you-go pricing model, where users pay only for the resources they use.